STANDARD OPENS: CONTAINMENT, FINITE REFINEMENT AND THE UNIT IDEAL CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 755-781, following the statement at 736-753.

Editorial note. Parts (2) and (3) are each a one-step consumption of a named cross-chapter algebra engine, quoted with the engine name; part (1) is argued explicitly.

Allowed standard prerequisites (editorial).

- Standard opens as spectra of localizations. Algebra, Lemma algebra-lemma-standard-open (cross-chapter, cited at ~757). D(g) = Spec(R_g).
- Zariski topology on Spec. Algebra, Lemma algebra-lemma-Zariski-topology (cross-chapter, cited at ~761, ~780). Nonvanishing on all primes => unit; unit ideal generation criterion.
- Quasi-compactness of standard opens. Algebra, Lemma algebra-lemma-qc-open (cross-chapter, cited at ~773). D(f) is quasi-compact, enabling finite refinement.

Source reference. Original reference 00E0: focused-seeds/geometry/S9-schemes-standard-open/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00F6: focused-seeds/geometry/S9-schemes-standard-open/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00E4: focused-seeds/geometry/S9-schemes-standard-open/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a ring. Let $f \in R$.*

(1) *If $g \in R$ and $D(g) \subset D(f)$, then*
   (a) *f is invertible in R_g,*
   (b) *$g^e = af$ for some $e \geq 1$ and $a \in R$,*
   (c) *there is a canonical ring map $R_f \to R_g$, and*
   (d) *there is a canonical R_f-module map $M_f \to M_g$ for any R-module M.*

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *Any open covering of $D(f)$ can be refined to a finite open covering of the form $D(f) = \bigcup_{i=1}^{n} D(g_i)$.*
(3) *If $g_1, \ldots, g_n \in R$, then $D(f) \subset \bigcup D(g_i)$ if and only if $g_1, \ldots, g_n$ generate the unit ideal in R_f.*

Proof. Recall that $D(g) = \mathrm{Spec}(R_g)$ (see Algebra, Lemma 00E4). Thus (a) holds because f maps to an element of R_g which is not contained in any prime ideal, and hence invertible, see Algebra, Lemma 00E0. Write the inverse of f in R_g as a/g^d. This means $g^d - af$ is annihilated by a power of g, whence (b). For (c), the map $R_f \to R_g$ exists by (a) from the universal property of localization, or we can define it by mapping b/f^n to $a^n b/g^{ne}$. The equality $M_f = M \otimes_R R_f$ can be used to obtain the map on modules, or we can define $M_f \to M_g$ by mapping x/f^n to $a^n x/g^{ne}$.

Recall that $D(f)$ is quasi-compact, see Algebra, Lemma 00F6. Hence the second statement follows directly from the fact that the standard opens form a basis for the topology.

The third statement follows directly from Algebra, Lemma 00E0. □